

# Analysis 2 — How Diverse Is a Country’s Cinema?

Genre diversity (Shannon entropy) across 8 major film-producing countries — and whether it tracks the harvest

## 1. The questions

After Analysis 1 found that *individual* genres don’t respond to crop yields, Analysis 2 steps back and asks about the **whole genre mix** at once:

- **(A) Descriptive:** Which countries make the most genre-diverse cinema, what drives those differences, and is diversity changing over time?
- **(B) Crop link:** Do countries make *more diverse* films after good harvests?

**Diversity metric — Shannon entropy.** For each country-year we take every (film, genre) tag, compute each genre’s share  $p_i$ , and calculate

$$H = - \sum p_i \cdot \ln(p_i)$$

Higher H = output spread evenly across many genres; lower H = output concentrated in a few. We also report **evenness** ( $H \div \ln(\text{richness})$ ), which removes the effect of *how many* genres exist) and **richness** (the count of distinct genres).

## 2. Method

Same data hygiene as Analysis 1:

- Films from `df_genre` — domestic productions ( $\leq 3$  distribution regions) with  $\geq 50$  votes.
- Country-years with **< 20 films dropped** (entropy is unstable on small slates).
- 8 countries clear the bar with  $\geq 5$  usable years: US, Japan, Germany, France, India, Brazil, Mexico, China. (South Korea and Nigeria too thin.)
- Crop test uses each country’s hand-picked staple yield as **year-over-year % growth** (detrended), correlated against the **year-over-year change in entropy**, same-year and at a 2-year lag, then aggregated across countries — exactly the method trusted in A1.

## 3. Results — descriptive

### 3a. The diversity ranking

| Rank | Country | Entropy | Evenness | Richness | Years |
|------|---------|---------|----------|----------|-------|
| 1    | Japan   | 2.35    | 0.85     | 16.0     | 30    |

| Rank | Country       | Entropy | Evenness | Richness | Years |
|------|---------------|---------|----------|----------|-------|
| 2    | United States | 2.33    | 0.81     | 18.1     | 72    |
| 3    | Germany       | 2.29    | 0.81     | 17.1     | 34    |
| 4    | Mexico        | 2.19    | 0.84     | 13.8     | 16    |
| 5    | Brazil        | 2.18    | 0.82     | 14.7     | 33    |
| 6    | India         | 2.06    | 0.81     | 13.1     | 43    |
| 7    | France        | 1.97    | 0.78     | 13.0     | 48    |
| 8    | China         | 1.86    | 0.77     | 11.5     | 15    |

**Japan, the United States, and Germany** (effectively tied at the top) produce the most genre-balanced cinema; **France and China** the most concentrated. The spread is modest (1.86–2.35) — every industry is reasonably broad — but the ends are meaningful.

### Cinematic Genre Diversity by Country

Shannon entropy of the genre mix (avg across years) — 8 major film producers

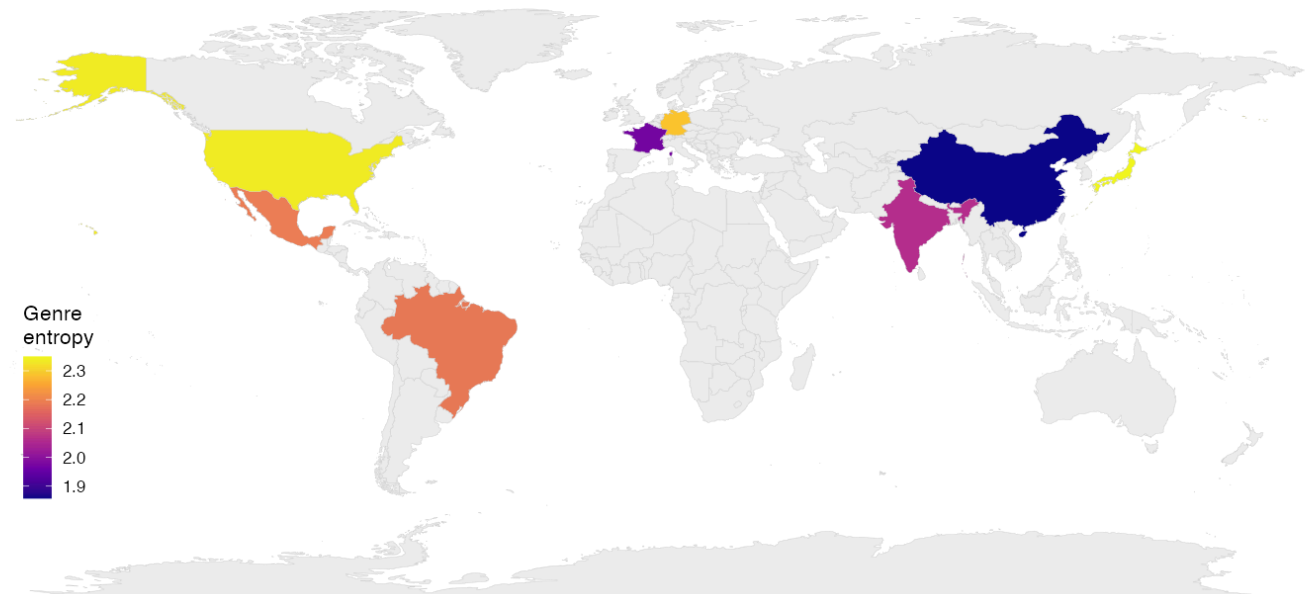


Figure 1. Average genre diversity by country (8 producers shaded; others have no data).

## Genre Diversity by Country

Average Shannon entropy of the genre mix

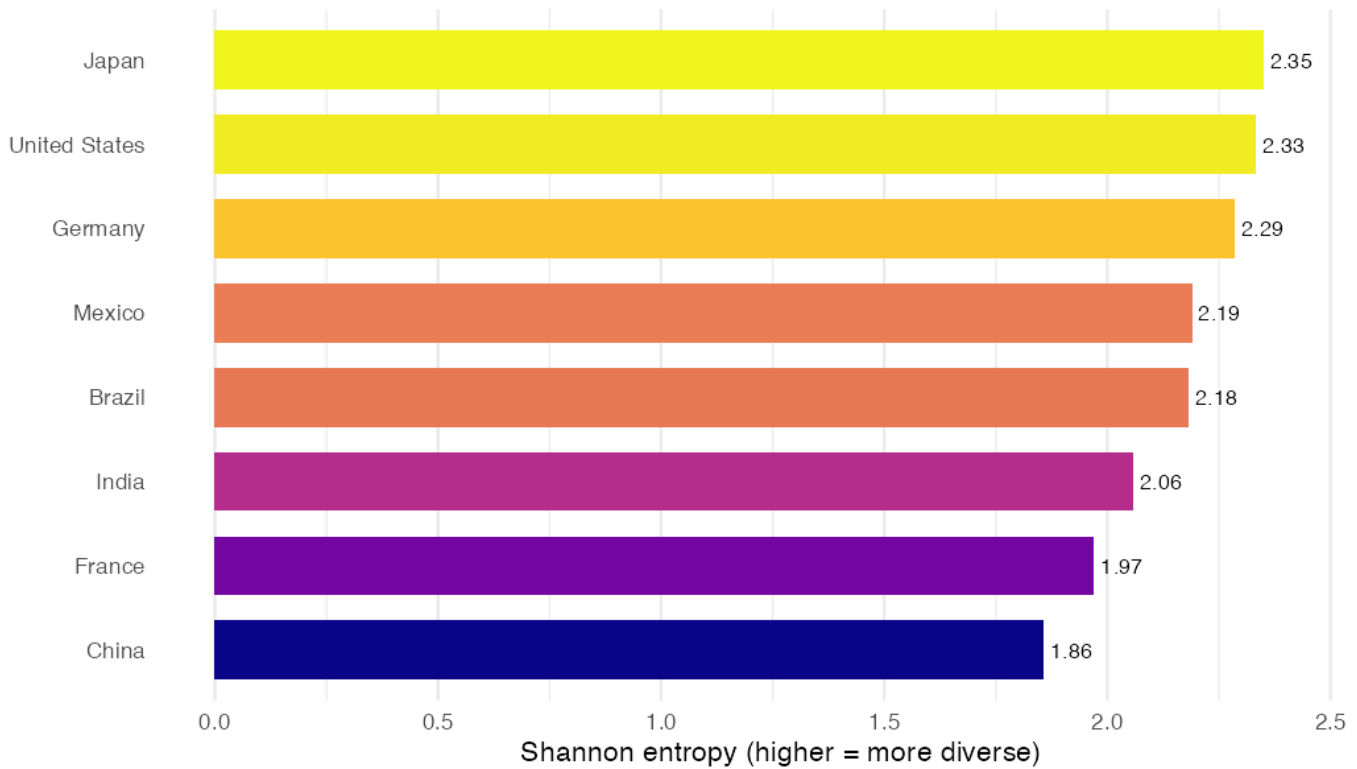


Figure 2. The precise ranking behind the map.

### 3b. Why? Every cinema is Drama-led — they differ in how hard they lean in

The single most common genre is **Drama in all 8 countries**. What differs is *how dominant* it is — and that lines up almost exactly with the diversity ranking:

| Country       | Drama share | Diversity |
|---------------|-------------|-----------|
| India         | 32.5%       | low       |
| France        | 30.0%       | low       |
| China         | 29.6%       | lowest    |
| Japan         | 28.0%       | highest   |
| Brazil        | 27.6%       | mid       |
| Mexico        | 26.2%       | mid       |
| Germany       | 24.3%       | high      |
| United States | 21.7%       | high      |

Low-diversity countries concentrate ~30–33% of their output in Drama; high-diversity ones hold Drama to ~22–24% and spread the rest across many genres. (India being the *most* Drama-concentrated matches Bollywood’s well-known lean toward drama/romance.) **Japan** is the instructive exception: its Drama share is mid-pack, but it distributes everything else so evenly (highest evenness, 0.85) that it tops the ranking.

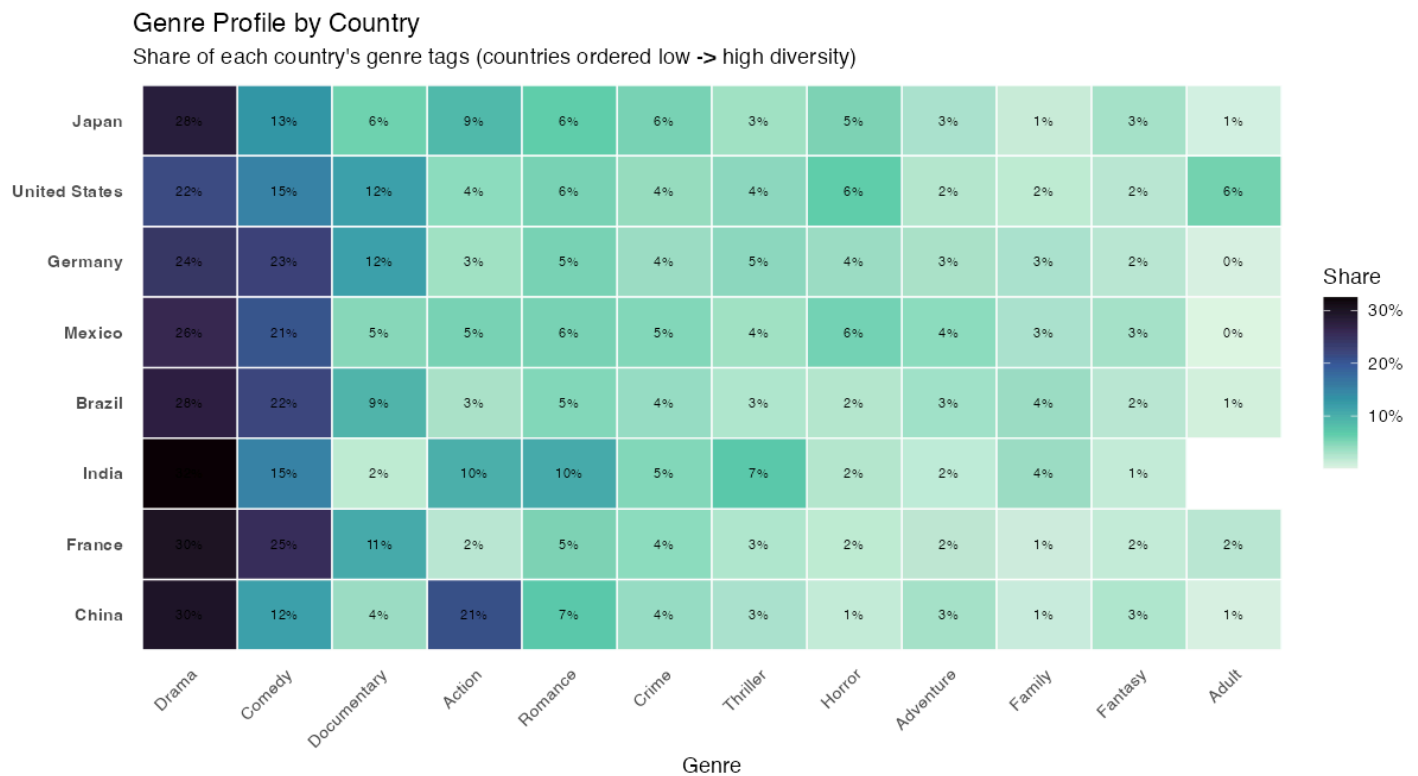


Figure 3. Each country's genre "fingerprint" — share of genre tags, countries ordered low→high diversity. The concentration into Drama (and a few neighbors) at the bottom vs. the even spread at the top is the visual story.

### 3c. Is this just “bigger industries make more genres”? Partly.

Diversity is **moderately correlated with film volume** (entropy vs. number of films:  $r = 0.45$  per country-year, 0.53 across countries). So part of the ranking does reflect that larger slates mechanically show more genres. But it is **not only** volume:

- The **US makes the most films yet ranks #2**, behind Japan — pure volume would put it first.
- **Evenness**, which is mathematically independent of how many genres appear, gives the same qualitative ordering (Japan/Mexico most even; China/France least).

So the ranking is part scale, part genuine composition — and the composition differences survive once scale is removed.

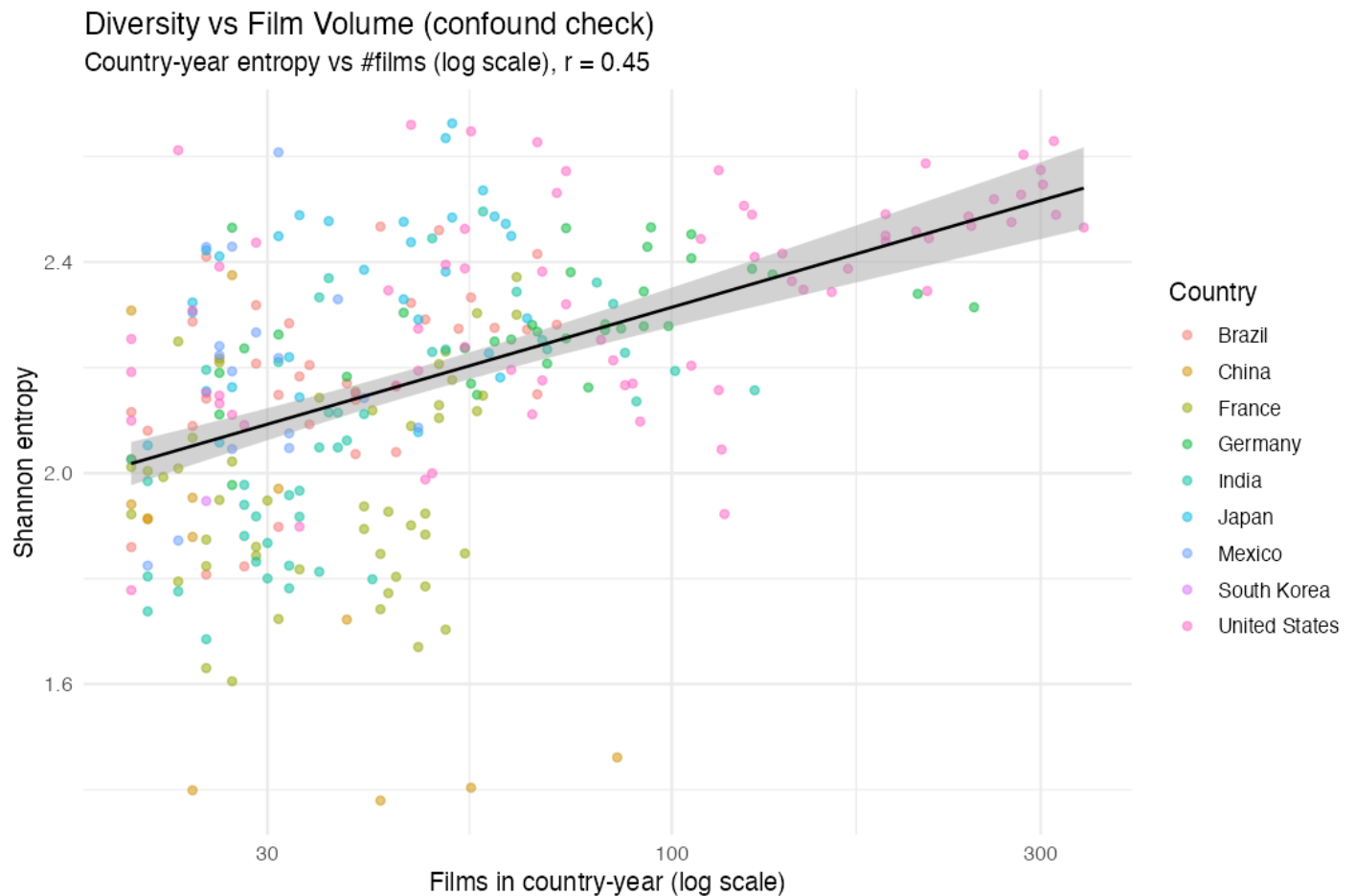


Figure 4. The confound check: entropy rises with film count ( $r = 0.45$ ), but loosely — volume explains only ~20% of the variation.

### 3d. Cinema is getting more diverse over time

Fitting entropy against year for each country, **7 of 8 show rising diversity** (~+0.05 to +0.10 entropy per decade), consistent with maturing, globalizing film industries. India rose fastest (+0.10/decade).

| Trend (entropy / decade) | Countries                                                                                                 |
|--------------------------|-----------------------------------------------------------------------------------------------------------|
| Rising                   | India (+0.10), Germany (+0.06), Brazil (+0.06), Japan (+0.06), US (+0.05), France (+0.04), Mexico (+0.02) |
| Falling                  | China (−0.19)                                                                                             |

China's decline is the lone exception, but rests on only 15 recent years — likely noise rather than a real reversal, and flagged as such.

## Genre Diversity Over Time

Shannon entropy of each country's genre mix, by year

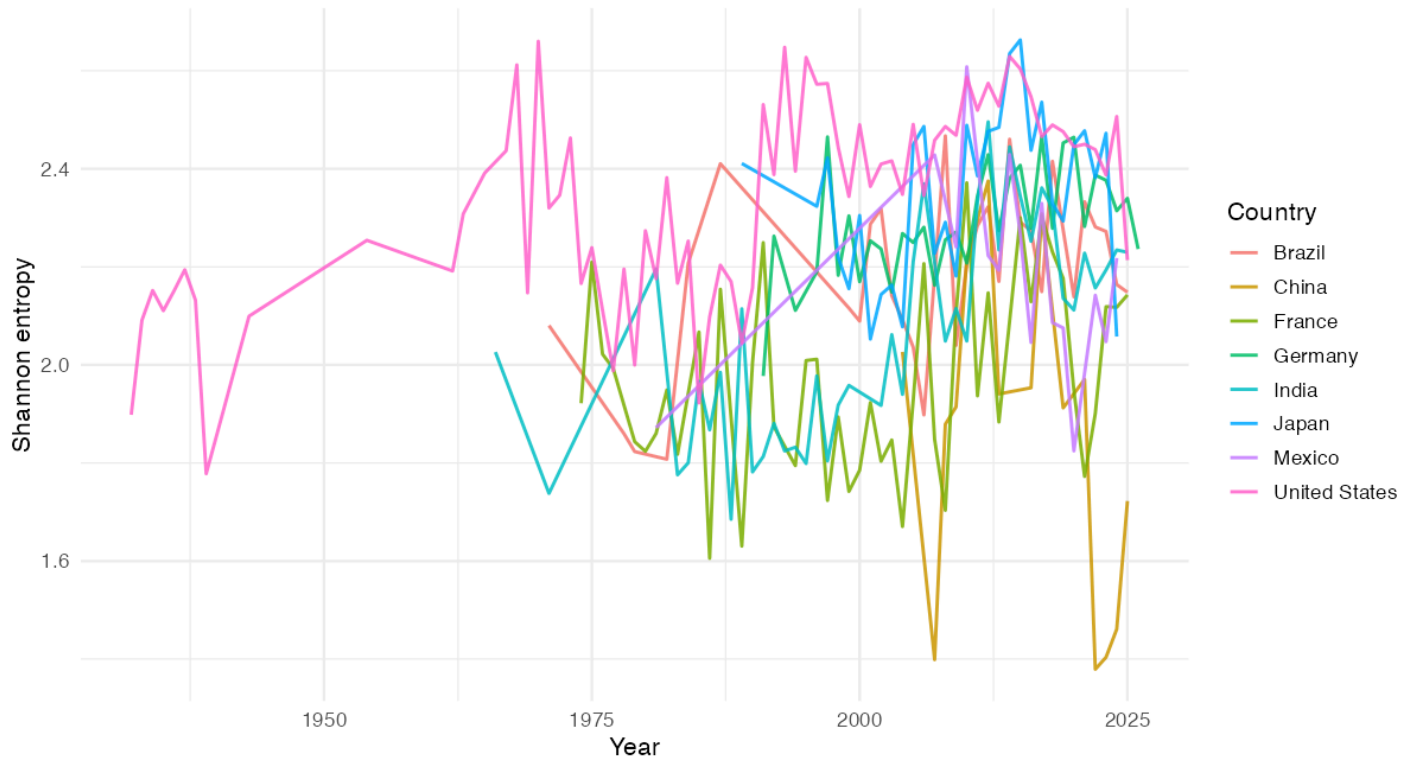


Figure 5. Genre entropy by year and country.

## 4. Results — the crop link (null)

Correlating year-over-year **change in diversity** against **staple-crop yield growth**, then averaging across the 8 countries:

| Lag              | mean r | sign split | p    |
|------------------|--------|------------|------|
| Same year        | +0.04  | 5 + / 3 –  | 0.59 |
| Harvest leads 2y | –0.05  | 3 + / 5 –  | 0.44 |

Both averages are essentially zero, neither is significant, and the sign flips between lags. Individual countries scatter inconsistently (e.g. Brazil +0.25 same-year but –0.35 at the 2-year lag). **Genre diversity does not track the harvest** — the same clean null as Analysis 1.

## Genre Diversity Change vs Staple Crop Yield Growth

Same year | each point = one year

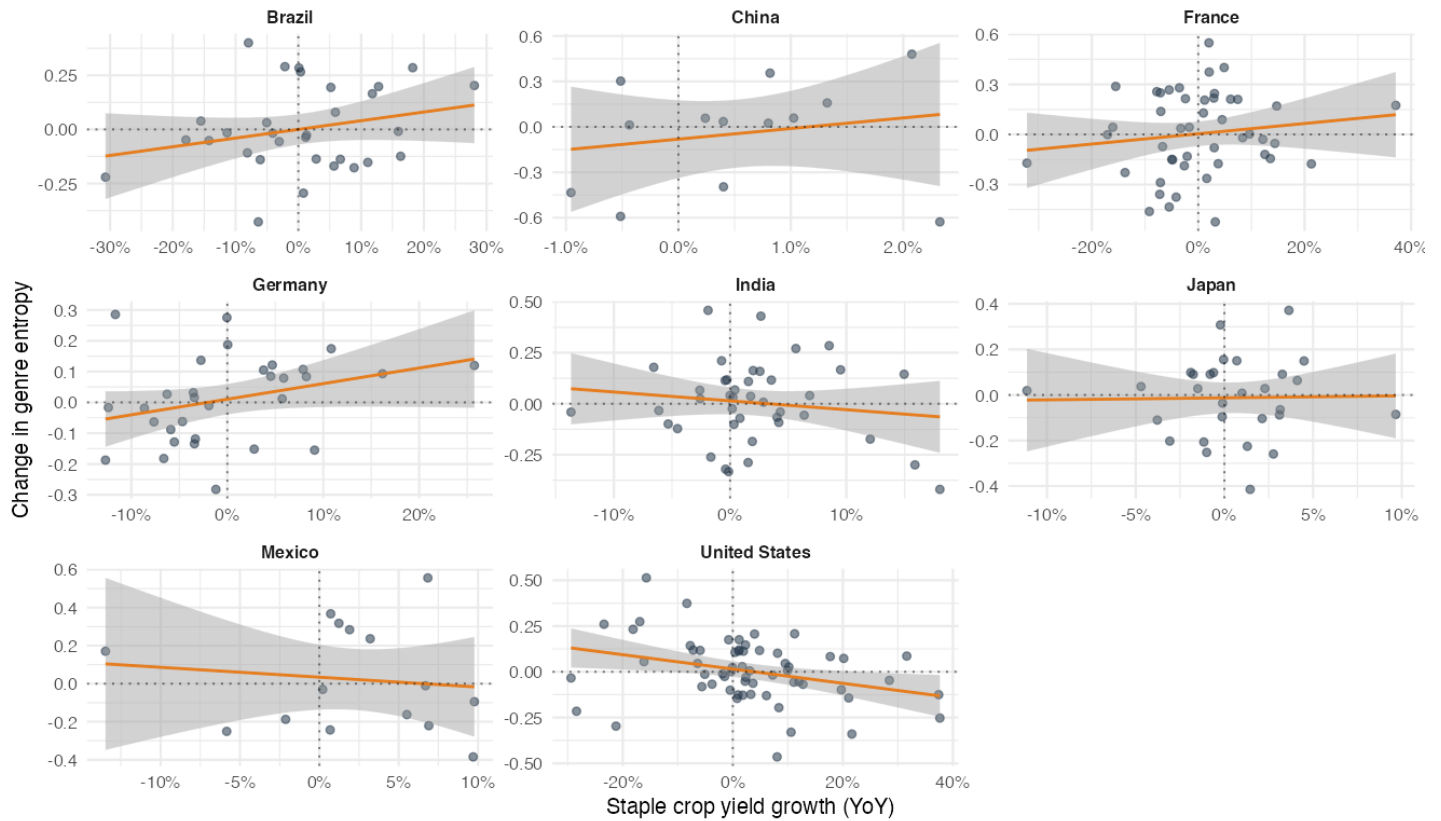


Figure 6. Year-over-year diversity change vs. staple yield growth, by country. Flat, inconsistent slopes.

## 5. Conclusion

- **Genre diversity differs meaningfully across countries** — Japan, the US, and Germany are the most genre-balanced; France and China the most concentrated.
- **The driver is how hard a country leans into Drama:** every cinema is Drama-led, but low-diversity countries pour  $\sim 1/3$  of output into it while high-diversity ones cap it near  $1/3$  and spread the rest.
- **Diversity has risen over time** in nearly every country.
- Part of the ranking reflects industry **scale** ( $r \approx 0.5$  with film volume), but the compositional differences are real and survive an evenness adjustment.
- **Crop yields have no detectable relationship with genre diversity** — reinforcing Analysis 1's finding that agriculture and cinema move independently.

Together with Analysis 1, the project tested the crop→cinema idea from two angles (single genres and overall diversity) and found no link either way — while still surfacing a genuine, well-explained pattern in *how* national cinemas differ.

## 6. Limitations

- **Film-count sensitivity.** Entropy correlates with volume ( $r \approx 0.5$ ); evenness is the more scale-robust read and is reported alongside it.
- **Thin samples.** China (15 yrs) and Mexico (16 yrs) give noisier estimates; China's negative time trend in particular should not be over-read.

- **IMDB genre tags** are crowd-sourced and US-centric, and the  $\leq 3$ -region origin proxy biases samples toward smaller domestic releases — both may compress measured diversity.
- **Only the same 8 countries** clear the data thresholds, so the map is sparse.